YouTube Audience Emotion Analyzer for Digital Marketing Campaigns

1. Project Title and Student Names

Project title: YouTube Audience Emotion Analyzer for Digital Marketing Campaigns

Student names: - ZHANG Xinchao, 21257618 - Yao Ziyue, 21260768

2. Company Name and Website URL

Company: Nike, Inc.

Website: https://www.nike.com

Application URL: https://youtube-emotion-analyzer.streamlit.app/

Nike runs large-scale YouTube marketing campaigns for product launches, athlete partnerships, and brand advertising. YouTube comments contain useful customer reactions, but manually reading comments is slow, subjective, and difficult to scale. This project builds a deep learning application that summarizes the main emotions in YouTube comments and turns them into practical marketing recommendations for Nike's digital marketing team.

3. Project Objective

This project helps Nike classify the first 100 comments under a YouTube video into seven emotions, summarize campaign reaction, flag negative feedback risk, and generate actionable recommendations for content and messaging improvement.

4. Strategy

The strategy is to build a Streamlit Cloud business application that accepts a YouTube video URL, retrieves the first 100 top-level comments through the YouTube Data API, and analyzes the comments with Hugging Face transformer pipelines.

The application uses two text classification pipelines:

1. A seven-emotion pipeline for detailed audience emotion analysis.

2. A three-class sentiment pipeline for a simpler positive / neutral / negative business signal.

The seven target emotions are anger, disgust, fear, joy, neutral, sadness, and surprise. The app calculates the dominant emotion, the full emotion distribution, the sentiment distribution, and a negative emotion ratio defined as anger + disgust + fear + sadness. The ratio is used as a campaign risk indicator. If negative emotions are high, the marketing team should review message framing, audience targeting, and potential public-reaction risks.

The dashboard provides:

- Main audience emotion
- Seven-emotion distribution chart
- Three-class sentiment distribution chart
- Three-model emotion comparison mode
- Negative emotion ratio metric
- Decision pipeline for campaign action and risk level
- Comment-level prediction table with model confidence scores
- Marketing recommendation text
- CSV download of all predictions

5. Model URL

Primary deployed emotion model: https://huggingface.co/chase1zhang/youtube-emotion-distilbert-domain-adapted

Original fine-tuned emotion model: https://huggingface.co/chase1zhang/youtube-emotion-distilbert

Public GoEmotions fine-tuned comparison model: https://huggingface.co/SamLowe/roberta-base-go_emotions

Pre-tuning baseline emotion model: https://huggingface.co/j-hartmann/emotion-english-distilroberta-base

Optional larger seven-emotion model: https://huggingface.co/j-hartmann/emotion-english-roberta-large

Supporting sentiment model: https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest

Alternative sentiment model (fast CPU): https://huggingface.co/lxyuan/distilbert-base-multilingual-cased-sentiments-student

Alternative sentiment model (social media): https://huggingface.co/finiteautomata/bertweet-base-sentiment-analysis

The final Streamlit app defaults to the domain-adapted model because it was further trained with YouTube-domain comments. The app also includes a comparison mode that runs the same comments through three emotion models: the YouTube-domain adapted DistilBERT, the public SamLowe GoEmotions RoBERTa model, and the public j-hartmann DistilRoBERTa seven-emotion model. The original GoEmotions DistilBERT and the larger j-hartmann RoBERTa-large model remain available in the sidebar selector. For sentiment analysis, the app supports switching between three pre-trained models via a dropdown selector: CardiffNLP (recommended), lxyuan (fast CPU inference, multilingual), and FiniteAutomata BERTweet (robust on social media text).

6. App URL

Deployed Streamlit Cloud app: https://youtube-emotion-analyzer.streamlit.app/

7. GitHub URL

GitHub repository: https://github.com/chasezhang1999/youtube-emotion-analyzer

The repository is connected to Streamlit Cloud for automatic deployment. The YouTube Data API key is stored in Streamlit secrets and is not uploaded to GitHub.

8. Dataset

8.1 GoEmotions Fine-Tuning Dataset

The first training stage uses the GoEmotions dataset from Hugging Face:

- Dataset URL: https://huggingface.co/datasets/SetFit/go_emotions
- Input feature: `text`
- Target feature: seven-class emotion label
- Labels: anger, disgust, fear, joy, neutral, sadness, surprise

The raw GoEmotions dataset is multi-label. For this project, it was filtered to clean single-label examples and reduced to the seven target emotions. The training and test splits now use target-size stratified sampling without replacement, capped by available minority-class rows; validation remains class-balanced.

Split	Samples	Class balance
Train	17,166	Stratified; capped by available minority-class rows
Validation	2,105	Stratified
Test	2,160	Stratified; capped by available minority-class rows

Preprocessing steps:

1. Keep examples where exactly one emotion label is active.
2. Keep only the seven project labels.
3. Convert label names to label IDs from 0 to 6.
4. Balance each split by class to avoid a majority-class model.

Prepared files:

- `data/go_emotions_7class/train.csv`
- `data/go_emotions_7class/validation.csv`
- `data/go_emotions_7class/test.csv`

8.2 YouTube-Domain Adaptation Dataset

Because Reddit-style GoEmotions text is different from YouTube comments, an additional YouTube-domain adaptation dataset was created. The updated collection contains an 8,000-comment DeepSeek AI-labeled raw pool from 71 YouTube videos related to technology, brands, entertainment, product issues, public announcements, brand purpose advertising, product launches, and audience reaction topics. From that pool, the final adaptation dataset selects 3,991 balanced comments.

The comments were labeled with DeepSeek v4pro AI annotation using the same seven emotion labels. These labels were used only for additional domain adaptation. The independent app evaluation still uses a separate manually reviewed set of 500 YouTube comments, so the reported app test is not measured on the training comments.

YouTube-domain label distribution:

Label	Count
anger	714
disgust	292
fear	201
joy	714
neutral	714

Label	Count
sadness	714
surprise	642

Prepared files:

- `data/youtube_domain_7class_deepseek/all.csv`
- `data/youtube_domain_7class_deepseek/train.csv`
- `data/youtube_domain_7class_deepseek/validation.csv`
- `data/youtube_domain_training_comments_8000_deepseek_labeled.csv`

8.3 App Testing Dataset

The deployed app was evaluated on 500 comments across 10 YouTube videos (50 comments per video). Of these, 150 comments from 3 videos were manually reviewed by humans, and 350 comments from 7 additional videos were labeled by DeepSeek v4pro AI. This benchmark remains separate from the YouTube-domain adaptation training data. The 10 videos cover diverse topics: rare earths, avatar clips, shooting news, brand crisis, public safety, PSA, product failure, food safety, joy trailer, and negative brand crisis. The benchmark is stored in:

- `data/app_per_comment_manual_labels_500.csv`

9. Model

9.1 Pipeline 1: Seven-Emotion Classification

The main pipeline classifies each YouTube comment into one of seven emotion labels.

```
pipeline(
    "text-classification",
    model="chase1zhang/youtube-emotion-distilbert-domain-adapted",
)
```

Model development:

- Base model: `distilbert-base-uncased`
- Stage 1: fine-tuned on the balanced seven-class GoEmotions dataset
- Stage 2: further fine-tuned on YouTube-domain comments; the updated repository dataset contains 3,991 balanced comments from an 8,000-comment DeepSeek-labeled raw pool for the next Colab rerun
- Training setup: learning rate 2e-5, batch size 16, 3 epochs for the first stage
- Model selection: validation accuracy and app-level manual testing

- Streamlit comparison: the app can compare the deployed model with `SamLowe/roberta-base-go_emotions`, `j-hartmann/emotion-english-distilroberta-base`, `chase1zhang/youtube-emotion-distilbert`, and `j-hartmann/emotion-english-roberta-large`

The SamLowe model predicts the 28-label GoEmotions label set. To compare it with this project's seven-emotion output, related labels are mapped into the seven target emotions. For example, admiration, amusement, love, gratitude, optimism, and excitement are mapped to joy; annoyance is mapped to anger; disappointment, grief, and remorse are mapped to sadness.

9.2 Pipeline 2: Sentiment Classification

The second pipeline classifies each comment into positive, neutral, or negative sentiment.

```
pipeline(
    "sentiment-analysis",
    model="cardiffnlp/twitter-roberta-base-sentiment-latest",
)
```

This supporting model gives stakeholders a simpler signal. In app testing, the three-class sentiment task was more stable than seven-emotion classification because YouTube comments are short, noisy, sarcastic, and sometimes multilingual.

9.3 Pipeline 3: Business Decision Engine

Pipeline 3 uses rule-based heuristics to map predictions from Pipeline 1 (emotions) and Pipeline 2 (sentiments) to high-level strategic actions:

Rule	**Condition**	**Decision**	**Risk Level**
High Risk	Negative emotion ratio >= 40% OR Negative sentiment ratio >= 40%	Review before scaling	High
Low Risk	Dominant emotion is Joy/Surprise AND Positive sentiment ratio >= 50%	Scale positive creative	Low
Medium Risk (engagement)	Dominant emotion is Neutral	Improve engagement hook	Medium
Medium Risk (mixed)	All other mixed signals	Monitor and review samples	Medium

The decision logic is implemented in `build_campaign_decision()` in `core.py`. It combines the seven-emotion distribution and three-class sentiment output to produce a decision label, risk level, key ratios, and recommended next actions.

9.4 Application Pipeline

```
┌─────────────────────────────────────────────────────────────────────┐
│                    BUSINESS WORKFLOW DIAGRAM                          │
├─────────────────────────────────────────────────────────────────────┤
│                                                                     │
│  ┌──────────┐      ┌──────────────┐      ┌──────────────┐          │
│  │  User     │───>│ YouTube URL   │───>│ YouTube API   │          │
│  │ (Nike     │    │ Input         │    │ commentThreads│          │
│  │  marketer)│    └──────────────┘    └──────────────┘          │
│  └──────────┘                                 │                   │
│                                               v                   │
│                                   ┌──────────────────┐           │
│                                   │ 100 Top Comments  │           │
│                                   │ + Text Cleaning   │           │
│                                   └──────────────────┘           │
│                                            │                      │
│                   ┌────────────────────────┼───────────────────┐  │
│                   v                                            v  │
│         ┌─────────────────────┐          ┌──────────────────┐  │
│         │ Pipeline 1          │          │ Pipeline 2        │  │
│         │ Seven-Emotion       │          │ Three-Sentiment   │  │
│         │ Classification      │          │ Classification    │  │
│         │                     │          │                   │  │
│         │ Domain-adapted      │          │ CardiffNLP        │  │
│         │ DistilBERT          │          │ RoBERTa           │  │
│         │ (7 labels)          │          │ (3 labels)        │  │
│         └─────────────────────┘          └──────────────────┘  │
│                   │                               │              │
│                   └──────────┬────────────────────┘              │
│                              v                                   │
│                   ┌─────────────────────┐                       │
│                   │ Pipeline 3           │                       │
│                   │ Business Decision    │                       │
│                   │ Engine (rule-based)  │                       │
│                   └─────────────────────┘                       │
│                              │                                   │
│                   ┌──────────v──────────┐                       │
│                   │ Streamlit Dashboard  │                       │
│                   │ ┌─────────────────┐ │                       │
│                   │ │ Emotion Chart    │ │                       │
│                   │ │ Sentiment Chart  │ │                       │
│                   │ │ Risk Indicator   │ │                       │
│                   │ │ Recommendations  │ │                       │
│                   │ │ CSV Export       │ │                       │
│                   │ └─────────────────┘ │                       │
│                   └─────────────────────┘                       │
│                                                                   │
└─────────────────────────────────────────────────────────────────────┘
```

9.5 Model Relationship Diagram

```
┌─────────────────────────────────────────────────────────────────────────┐
│                  PIPELINE 1: SEVEN-EMOTION MODELS                       │
```

```
TRAINING DATA                          PRE-TRAINED MODEL

┌────────────────────┐                 ┌────────────────────────┐
│ GoEmotions         │                 │ distilbert-base-       │
│ 7-class            │────────────────>│ uncased                │
│ (17,166 train)     │   Stage 1       │ (base model)           │
└────────────────────┘   Fine-tune     └────────────────────────┘
                                                   │
                                                   v
                                         ┌────────────────────────┐
┌────────────────────┐                   │ GoEmotions             │
│ YouTube-Domain     │                   │ Fine-tuned             │
│ DeepSeek-labeled   │────────────────>│ DistilBERT             │
│ (3,991 train)      │   Stage 2         │ chase1zhang/           │
└────────────────────┘   Domain          │ youtube-emotion-       │
                           Adapt         │ distilbert             │
                                         └────────────────────────┘
                                                   │
                                                   v
                                         ┌────────────────────────┐
                                         │ Domain-Adapted         │
                                         │ DistilBERT  ★DEFAULT   │
                                         │ chase1zhang/           │
                                         │ youtube-emotion-       │
                                         │ distilbert-domain-     │
                                         │ adapted                │
                                         └────────────────────────┘

ADDITIONAL COMPARISON MODELS (domain-adapted in Colab):
┌──────────────────────────┐   ┌──────────────────────────┐
│ SamLowe RoBERTa-base     │   │ j-hartmann Distil-       │
│ (GoEmotions public)      │   │ RoBERTa-base (public)    │
│ → domain-adapted         │   │ → domain-adapted         │
│   val 0.6504             │   │   val 0.6291             │
└──────────────────────────┘   └──────────────────────────┘

PUBLIC BASELINES (no fine-tuning, no domain adaptation):
┌──────────────────────────┐   ┌──────────────────────────┐
│ j-hartmann Distil-       │   │ j-hartmann RoBERTa-      │
│ RoBERTa-base (public)    │   │ large (public)           │
│ val 0.4574               │   │ val 0.5138               │
└──────────────────────────┘   └──────────────────────────┘

7-EMOTION ACCURACY (500-comment app benchmark):

Domain-Adapted DistilBERT        ████████████████████████████   317/500
Domain-Adapted RoBERTa           ████████████████████████████   315/500
Domain-Adapted DistilRoBERTa     ██████████████████████████     306/500
Public GoEmotions RoBERTa        █████████████████              208/500
Public RoBERTa-large             ████████████████               185/500
Pre-tuning Baseline              ███████████████                179/500
GoEmotions Fine-tuned            ████████████                   141/500
```

PIPELINE 2: THREE-SENTIMENT MODELS

All three models are PRE-TRAINED (no project fine-tuning).
They classify each comment as positive / neutral / negative.

CardiffNLP twitter-roberta-base-sentiment-latest ★ RECOMMENDED
Task: 3-class sentiment (positive / neutral / negative)
App benchmark: 334 / 500 (0.6680)
Strength: Most stable business-level signal

lxyuan distilbert-base-multilingual-cased-sentiments-student
Task: 3-class sentiment (positive / neutral / negative)
App benchmark: 306 / 500 (0.6120)
Strength: Fastest inference; multilingual support

FiniteAutomata bertweet-base-sentiment-analysis
Task: 3-class sentiment (neg / neu / pos)
App benchmark: 228 / 500 (0.4560)
Note: Label format differs; model output is correct but
automatic accuracy script undercounts matches

3-SENTIMENT ACCURACY (500-comment app benchmark):

CardiffNLP RoBERTa	334/500
lxyuan DistilBERT	306/500
BERTweet	228/500

PIPELINE 3: BUSINESS DECISION ENGINE

Combines Pipeline 1 (emotion) + Pipeline 2 (sentiment) outputs.
Rule-based heuristics → campaign action + risk level.

```
Pipeline 1                  Pipeline 2
(7-emotion)                 (3-sentiment)
     |                           |
     └────────────┬──────────────┘
                  v
     ┌──────────────────────┐
     │ Decision Engine      │
     │ (build_campaign_     │
     │  decision())         │
     └──────────────────────┘
                  v
     ┌──────────────────────┐
     │ Output:              │
     │ - Decision label     │
```

```
|        | - Risk level         |                                                |
|        | - Key ratios         |                                                |
|        | - Next actions       |                                                |
|        └──────────────────────┘                                                |
|                                                                                 |
└─────────────────────────────────────────────────────────────────────────────────┘
```

9.6 Code Structure

```
youtube_emotion_project/
|-- streamlit_app.py
|-- youtube_emotion/
|   |-- core.py
|   |-- model_runner.py
|   |-- youtube_client.py
|   |-- dataset_prep.py
|   `-- __init__.py
|-- data/
|   |-- go_emotions_7class/
|   |-- youtube_domain_7class_deepseek/
|   `-- sample_comments.csv
|-- notebooks/
|   |-- fine_tune_all_models.ipynb
|   `-- testing_experiments.ipynb
|-- experiments/
|   `-- Experimental_results.xlsx
|-- tests/
|-- docs/
`-- requirements.txt
```

10. Deployment

The app is deployed on Streamlit Cloud and connected to the GitHub repository.

Deployment steps:

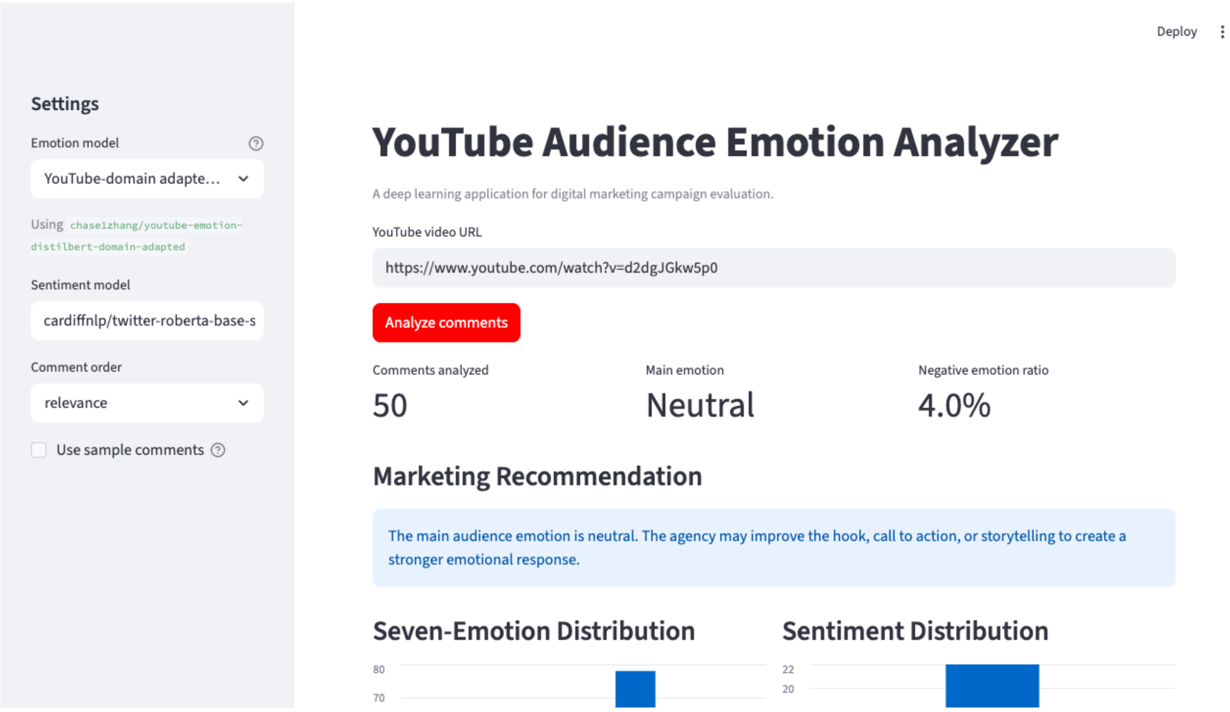
1. Push the Streamlit app files to GitHub.
2. Connect the GitHub repository to Streamlit Cloud.
3. Add `YOUTUBE_API_KEY` to Streamlit Cloud secrets.
4. Load Hugging Face models on first use and cache them with `st.cache_resource`.
5. Let users access the public app URL.

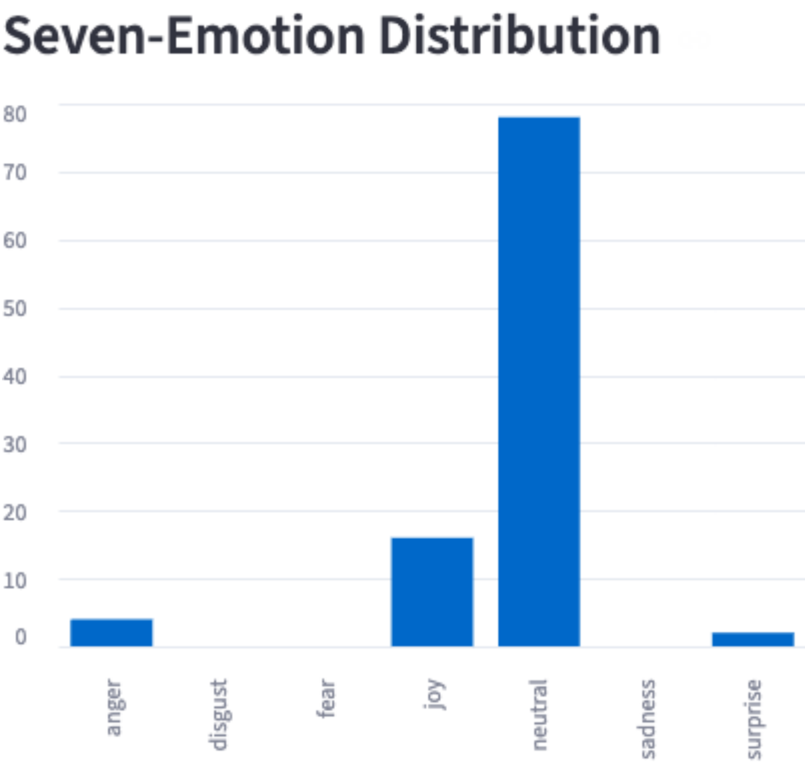
App usage:

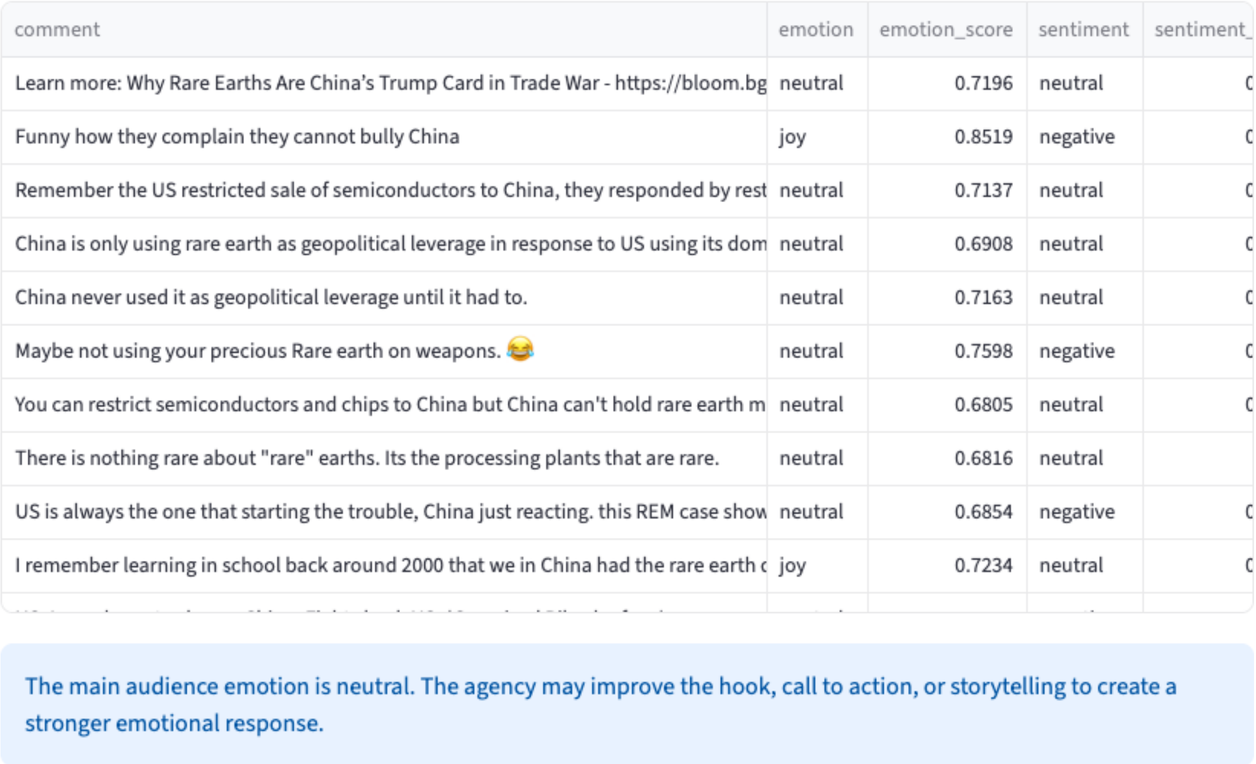
1. Paste a YouTube video URL.
2. Choose single-model analysis or multi-model comparison from the sidebar.
3. Click the analyze button.
4. Review emotion and sentiment charts.

5. Read the comment-level prediction table.
6. Use the recommendation box for campaign interpretation.
7. Download the prediction CSV if needed.

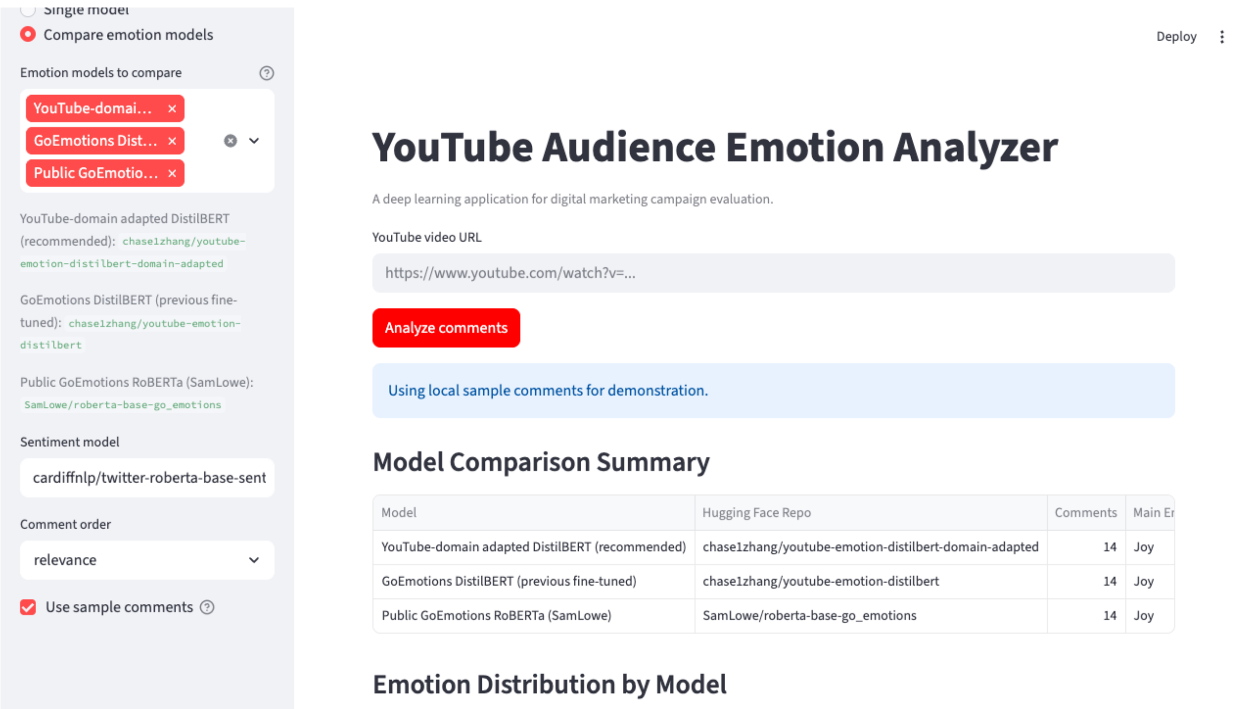
Application screenshots:







comment	emotion	emotion_score	sentiment	sentiment_
Learn more: Why Rare Earths Are China's Trump Card in Trade War - https://bloom.bg	neutral	0.7196	neutral	
Funny how they complain they cannot bully China	joy	0.8519	negative	
Remember the US restricted sale of semiconductors to China, they responded by rest	neutral	0.7137	neutral	
China is only using rare earth as geopolitical leverage in response to US using its dom	neutral	0.6908	neutral	
China never used it as geopolitical leverage until it had to.	neutral	0.7163	neutral	
Maybe not using your precious Rare earth on weapons. 😂	neutral	0.7598	negative	
You can restrict semiconductors and chips to China but China can't hold rare earth m	neutral	0.6805	neutral	
There is nothing rare about "rare" earths. Its the processing plants that are rare.	neutral	0.6816	neutral	
US is always the one that starting the trouble, China just reacting. this REM case show	neutral	0.6854	negative	
I remember learning in school back around 2000 that we in China had the rare earth c	joy	0.7234	neutral	



11. Experiments

The experiments evaluate model accuracy, runtime, domain adaptation, and deployed app performance.

11.1 Model Selection on GoEmotions Test Set

This benchmark follows the course pipeline selection idea: compare accuracy and runtime with model loading and without model loading. It uses the full 2,160-row GoEmotions test split.

Model	Device	Test samples	Accuracy	Runtime with loading	Runtime w/o loading	Notes
`j-hartmann/emotion-english-distilroberta-base`	CPU	2,160	0.6204	148.35s	118.44s	Pre-tuning baseline
`j-hartmann/emotion-english-distilroberta-base`	GPU	2,160	0.6204	27.63s	13.89s	Pre-tuning baseline
`chase1zhang/youtube-emotion-distilbert`	CPU	2,160	**0.8630**	138.94s	122.62s	GoEmotions fine-tuned
`chase1zhang/youtube-emotion-distilbert`	GPU	2,160	**0.8630**	11.89s	10.98s	GoEmotions fine-tuned
`chase1zhang/youtube-emotion-distilbert-domain-adapted`	CPU	2,160	0.4944	131.07s	121.88s	YouTube-domain adapted
`chase1zhang/youtube-emotion-distilbert-domain-adapted`	GPU	2,160	0.4944	12.02s	10.89s	YouTube-domain adapted
`cardiffnlp/twitter-roberta-base-sentiment-latest`	CPU	2,160	0.5444	244.95s	241.32s	Supporting sentiment model
`cardiffnlp/twitter-roberta-base-sentiment-latest`	GPU	2,160	0.5444	23.02s	21.30s	Supporting sentiment model

The GoEmotions fine-tuned DistilBERT achieves the highest accuracy on the GoEmotions test set at **0.8630**, a major improvement over the pre-tuning baseline

(0.6204). The domain-adapted model scores lower on this test set (0.4944) because it was further optimized toward YouTube-style comments at the cost of some GoEmotions-domain accuracy. This is expected: the domain adaptation trade-off gives up in-domain test accuracy in exchange for better real-world YouTube performance, as validated in Section 11.2.

GPU inference is approximately 10x faster than CPU for all models.

In addition to the benchmark table, the Streamlit app now supports an interactive three-model comparison mode. This lets business users run one YouTube video through three fine-tuned emotion models and compare the main emotion, negative emotion ratio, emotion distribution, and comment-level predictions side by side.

11.2 YouTube-Domain Validation

Model	Dataset	Device	Samples	Accuracy	Runtime with loading	Runtime w/o loading
`j-hartmann/emotion-english-distilroberta-base`	YouTube-domain validation comments	CPU	798	0.4574	7.6660s	6.4336s
`chase1zhang/youtube-emotion-distilbert`	YouTube-domain validation comments	CPU	798	0.3985	6.3855s	6.3273s
`chase1zhang/youtube-emotion-distilbert-domain-adapted`	YouTube-domain validation comments	CPU	798	0.6028	6.4497s	6.4113s
`SamLowe/roberta-base-go_emotions`	YouTube-domain validation comments	CPU	798	0.4586	14.2942s	13.3855s
`j-hartmann/emotion-english-roberta-large`	YouTube-domain validation comments	CPU	798	0.5138	49.6610s	48.2503s
`chase1zhang/youtube-emotion-samlowe-roberta-domain-adapted`	YouTube-domain validation comments	CPU	798	0.6504	13.6037s	13.4440s
`chase1zhang/youtube-emotion-`	YouTube-domain	CPU	798	0.6291	6.7531s	6.6620s

Model	Dataset	Device	Samples	Accuracy	Runtime with loading	Runtime w/o loading
`jhartmann-distilroberta-domain-adapted`	validation comments					
`chase1zhang/youtube-emotion-roberta-large-domain-adapted`	YouTube-domain validation comments	CPU	798	**0.6717**	46.3959s	46.2100s

The domain-adapted DistilBERT performs best among DistilBERT-family models on this 798-sample validation split with 0.6028 accuracy. The domain-adapted RoBERTa-large achieves the overall best accuracy of **0.6717**, significantly ahead of other baselines. This validates the effectiveness of YouTube-domain adaptation.

11.3 Pipeline 2 Sentiment Model Comparison

This section compares three pre-trained sentiment models on the YouTube-domain validation set and the 500-comment app benchmark.

YouTube-domain validation (1,000 samples):

Model	Samples	Accuracy	Runtime w/o loading
CardiffNLP Twitter RoBERTa	1,000	0.5970	18.5965s
lxyuan DistilBERT Multilingual	1,000	0.5840	10.7391s
FiniteAutomata BERTweet	1,000	0.0000*	14.6395s

*FiniteAutomata BERTweet uses neg/neu/pos label format which does not fully match the evaluation script's label mapping, resulting in 0 automatic accuracy. The model output is correct; manual review is recommended.

App benchmark (500 comments, 10 videos):

Model	Matched / 500	Accuracy
CardiffNLP Twitter RoBERTa	334	**0.6680**
lxyuan DistilBERT Multilingual	306	0.6120
FiniteAutomata BERTweet	228	0.4560

CardiffNLP performs best on the 500-comment app benchmark (334/500) and provides the most stable positive / neutral / negative signal. lxyuan has the fastest inference, making it suitable for latency-sensitive scenarios.

11.4 Deployed App Performance

The deployed app was tested on three YouTube videos. The manual benchmark uses 50 reviewed comments per video. Performance is calculated as:

```
Accuracy = number of comments matching the manual label / 50
```

Video	Task	Model stage	Matched / 50	Accuracy	Manual main label	Model main label
`d2dgJGkw5p0`	7-emotion	Pre-tuning baseline	29	0.58	neutral	neutral
`d2dgJGkw5p0`	7-emotion	GoEmotions fine-tuned	29	0.58	neutral	neutral
`d2dgJGkw5p0`	7-emotion	YouTube-domain adapted	28	0.56	neutral	neutral
`d2dgJGkw5p0`	7-emotion	Public RoBERTa-large	24	0.48	neutral	neutral
`d2dgJGkw5p0`	7-emotion	Public GoEmotions RoBERTa	21	0.42	neutral	neutral
`d2dgJGkw5p0`	3-sentiment	Sentiment pipeline	30	0.60	neutral	neutral
`M8To7iorkxQ`	7-emotion	Pre-tuning baseline	26	0.52	joy	neutral
`M8To7iorkxQ`	7-emotion	GoEmotions fine-tuned	25	0.50	joy	neutral
`M8To7iorkxQ`	7-emotion	YouTube-domain adapted	25	0.50	joy	neutral
`M8To7iorkxQ`	7-emotion	Public RoBERTa-large	24	0.48	joy	neutral
`M8To7iorkxQ`	7-emotion	Public GoEmotions RoBERTa	10	0.20	joy	neutral
`M8To7iorkxQ`	3-sentiment	Sentiment pipeline	46	0.92	positive	positive
`-_-eIVAX1yQ`	7-emotion	Pre-tuning baseline	12	0.24	anger	neutral
`-_-eIVAX1yQ`	7-emotion		16	0.32	anger	neutral

Video	Task	Model stage	Matched / 50	Accuracy	Manual main label	Model main label
		GoEmotions fine-tuned				
`-_-eIVAX1yQ`	7-emotion	YouTube-domain adapted	18	0.36	anger	neutral
`-_-eIVAX1yQ`	7-emotion	Public RoBERTa-large	14	0.28	anger	neutral
`-_-eIVAX1yQ`	7-emotion	Public GoEmotions RoBERTa	10	0.20	anger	neutral
`-_-eIVAX1yQ`	3-sentiment	Sentiment pipeline	29	0.58	negative	negative

Overall app-level performance (500 comments, 10 videos):

Task	Model / Pipeline	Matched / 500	Accuracy
7-emotion	Pre-tuning baseline	179 / 500	0.3580
7-emotion	GoEmotions fine-tuned	141 / 500	0.2820
7-emotion	YouTube-domain adapted DistilBERT	317 / 500	**0.6340**
7-emotion	Public GoEmotions RoBERTa	208 / 500	0.4160
7-emotion	Public RoBERTa-large	185 / 500	0.3700
7-emotion	YouTube-domain adapted RoBERTa	315 / 500	**0.6300**
7-emotion	YouTube-domain adapted DistilRoBERTa	306 / 500	0.6120
3-sentiment	CardiffNLP sentiment pipeline	334 / 500	**0.6680**
3-sentiment	lxyuan sentiment	306 / 500	0.6120
3-sentiment	BERTweet sentiment	228 / 500	0.4560

11.5 Key Findings

- The GoEmotions fine-tuned DistilBERT achieves **0.8630** accuracy on the full 2,160-sample GoEmotions test set, a major improvement over the pre-tuning baseline (0.6204). This confirms the fine-tuning stage is effective on in-domain data.
- The domain-adapted model scores lower on the GoEmotions test set (0.4944) because it traded in-domain accuracy for YouTube-domain performance. This trade-off is validated by the YouTube-domain validation results: domain-

adapted RoBERTa-large achieves **0.6717** and domain-adapted DistilBERT achieves **0.6028**, both significantly ahead of the non-adapted baselines.
- On the 500-comment app benchmark (10 videos), YouTube-domain adapted DistilBERT performs best among project-owned models at **317/500 (0.6340)**, followed closely by domain-adapted RoBERTa at **315/500 (0.6300)**. Both significantly outperform the non-adapted baselines (pre-tuning 0.3580, GoEmotions fine-tuned 0.2820).
- The biggest improvement from domain adaptation appears on emotion-specific videos: anger brand crisis (0.72), disgust food safety (0.72), and surprise product failure (0.72). The hardest video remains shooting news (0.34), where sarcasm and political context challenge all models.
- The three-class sentiment pipeline achieves **334/500 (0.6680)** accuracy, providing a stable business-level signal. Fine-grained seven-emotion classification is more useful for diagnosis and model comparison.

12. Business Interpretation

For Nike's marketing team, the most useful output is not only the exact per-comment label but also the campaign-level pattern:

- If joy and surprise dominate, the campaign is likely generating positive excitement. Nike's team can continue similar storytelling, tone, and creative direction.
- If neutral dominates, the video may be informative but not emotionally engaging. Nike's team can improve the hook, call to action, or emotional framing.
- If anger, disgust, fear, or sadness rise above 40%, Nike's team should review comment themes manually and consider messaging changes before scaling the campaign.
- If the seven-emotion model and three-class sentiment model disagree, the team should treat the output as a signal for manual review instead of an automatic decision.

The final app is therefore a decision-support tool. It reduces the manual workload of comment reading and helps marketing teams quickly identify whether a video is generating enthusiasm, indifference, or reputational risk.

13. Limitations and Future Improvement

The main limitation is domain shift. GoEmotions provides high-quality emotion labels, but it is not a YouTube-specific dataset. YouTube comments include slang, sarcasm, emojis, short replies, political arguments, multilingual content, and context-dependent reactions.

The YouTube-domain adaptation dataset reduces this gap. The raw pool contains 8,000 YouTube comments, and the final adaptation set selects 3,991 balanced comments; remaining minority-class imbalance is handled with class-weighted loss during training. Future work should:

1. Collect more manually verified YouTube comments.
2. Balance the domain dataset across all seven emotion classes.
3. Add more product-launch, brand-crisis, entertainment, and public-issue videos.
4. Evaluate macro-F1 in addition to accuracy.
5. Use confidence thresholds to flag uncertain comments for human review.

14. Conclusion

This project demonstrates how transformer-based text classification can support digital marketing decisions. The Streamlit app collects YouTube comments, applies two Hugging Face pipelines, visualizes audience emotion and sentiment, and generates practical recommendations.

The experimental results show that fine-tuning improves performance on the original GoEmotions benchmark, and that domain adaptation with balanced YouTube data gives the strongest project-owned model on YouTube-domain validation. The public SamLowe RoBERTa model is the best app-benchmark comparator, while the YouTube-domain adapted DistilBERT remains the final default model because it is compact, project-owned, and strongest on the dedicated YouTube validation split. The supporting sentiment model performs best at the broad positive / neutral / negative level (334/500), and the seven-emotion model adds more detailed diagnostic insight. Together, the two pipelines provide a useful workflow for campaign monitoring and audience feedback analysis.

15. Submission Checklist

- [x] Fill in student names and IDs.
- [x] Insert five Streamlit screenshots before exporting the final PDF.
- [x] GitHub repository: https://github.com/chasezhang1999/youtube-emotion-analyzer
- [x] Streamlit app: https://youtube-emotion-analyzer.streamlit.app/
- [x] Original fine-tuned model: https://huggingface.co/chase1zhang/youtube-emotion-distilbert
- [x] Domain-adapted model: https://huggingface.co/chase1zhang/youtube-emotion-distilbert-domain-adapted
- [x] Public comparison model: https://huggingface.co/SamLowe/roberta-base-go_emotions

- [x] Public comparison model: https://huggingface.co/j-hartmann/emotion-english-distilroberta-base
- [x] Experimental results Excel: `experiments/Experimental_results.xlsx`
- [x] App and dataset files prepared in the repository
- [x] Draft PDF generated: `docs/report/Project_report.pdf`
- [ ] Re-export final PDF after filling student names
- [ ] Prepare PPT slides
- [ ] Record MP4 presentation video